

Installing Git

Git runs on all operating systems, but there are different approaches to installing it on each system. The following sections provide specific instructions for each operating system.

Git is included on some systems by default, and is often bundled with other packages that you might have already installed. Before trying to install Git, see if it's already on your system. Open a new terminal window and issue the command `git --version`. If you see output listing a specific version number, Git is installed on your system. If you see a message prompting you to install or update Git, follow the onscreen instructions.

If you don't see any onscreen instructions and you're using Windows or macOS, you can download an installer from <https://git-scm.com>. If you're a Linux user with an apt-compatible system, you can install Git with the command `sudo apt install git`.

Configuring Git

Git keeps track of who makes changes to a project, even when only one person is working on the project. To do this, Git needs to know your username and email. You must provide a username, but you can make up a fake email address:

```
$ git config --global user.name "username"
$ git config --global user.email "username@example.com"
```

If you forget this step, Git will prompt you for this information when you make your first commit.

It's also best to set the default name for the main branch in each project. A good name for this branch is `main`:

```
$ git config --global init.defaultBranch main
```

This configuration means that each new project you use Git to manage will start out with a single branch of commits called *main*.

Making a Project

Let's make a project to work with. Create a folder somewhere on your system called *git_practice*. Inside the folder, make a simple Python program:

```
hello_git.py print("Hello Git world!")
```

We'll use this program to explore Git's basic functionality.

Ignoring Files

Files with the extension *.pyc* are automatically generated from *.py* files, so we don't need Git to keep track of them. These files are stored in a directory